Lab Assignment #11

Math 437 - Modern Data Analysis

Instructions

In this lab we will look at decision trees and ensembles of decision trees for both classification and regression.

```r
here::i_am("Labs/Lab11.qmd")

# You may not need all of this, but you'll need most of it
library(ISLR2)
library(dplyr)
library(rpart)
library(randomForest)
library(gbm)
library(tidymodels)
library(ranger)
library(vip)
library(xgboost)

# load and split public_2020 dataset
public <- readr::read_csv(here::here("Data/public_2020.csv"))
public <- public |>
  mutate(
    MSI = if_else(HBCU | HSI, "Yes", "No") |> as.factor()
  )

set.seed(3745)
public_split <- initial_split(
  public,
  prop = 0.75
)
public_train <- training(public_split)
public_test <- testing(public_split)
```

This lab assignment is worth a total of **25 points** with up to 5 points of extra credit.

Problem 1: Decision Trees

In past years, students have complained that the `tree` package wasn't running on their version of R. So instead of doing Labs 8.3.1 and 8.3.2 we will run similar code using the `rpart` package.

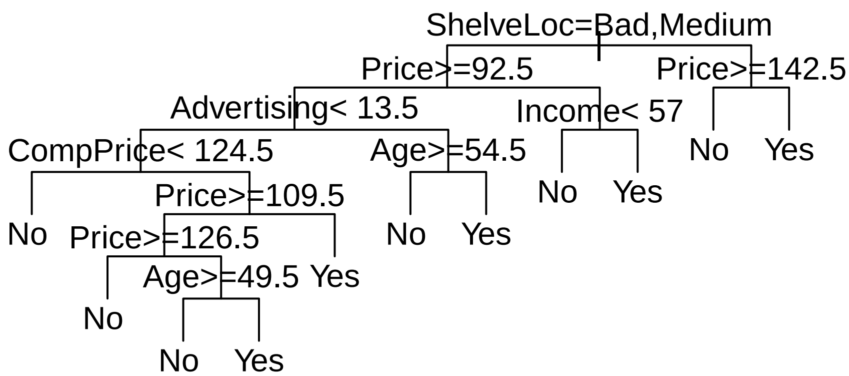
I have written each of the code chunks below to match Labs 8.3.1 and 8.3.2 as closely as possible. For each part, run the corresponding code chunks and answer the question about the output. You may find reading the text in Labs 8.3.1 and 8.3.2 to be useful.

Part a (Explanation: 1 pt)

```
Carseats2 <- Carseats |>
  mutate(
    High = if_else(Sales <= 8, "No", "Yes"))
```

```
tree.carseats <- rpart(High ~. - Sales, data = Carseats2)

par(xpd = TRUE)
plot(tree.carseats, uniform = TRUE)
text(tree.carseats, pretty = 0)
```



Briefly explain how to interpret this tree: which stores would end up in the very left "no" group, and which stores would end up in the very right "yes" group?

Stores that end up in the very left "no" group would be stores with bad shelf locations, prices that are less than 92.50 dollars, spend less than than 13,500 dollars on advertising,and their Competitor price is less than 124.50 dollars. Stores that end up in the very right "yes" group would be stores with good shelf locations and prices that are 142.50 dollar or more.

Solution

Part b (Explanation: 1 pt)

```
tree.carseats
```

```
n= 400

node), split, n, loss, yval, (yprob)
      * denotes terminal node

  1) root 400 164 No (0.59000000 0.41000000)
    2) ShelveLoc=Bad,Medium 315  98 No (0.68888889 0.31111111)
```

```
    4) Price>=92.5 269  66 No (0.75464684 0.24535316)
      8) Advertising< 13.5 224  41 No (0.81696429 0.18303571)
       16) CompPrice< 124.5 96   6 No (0.93750000 0.06250000) *
       17) CompPrice>=124.5 128  35 No (0.72656250 0.27343750)
         34) Price>=109.5 107  20 No (0.81308411 0.18691589)
           68) Price>=126.5 65   6 No (0.90769231 0.09230769) *
           69) Price< 126.5 42  14 No (0.66666667 0.33333333)
            138) Age>=49.5 22   2 No (0.90909091 0.09090909) *
            139) Age< 49.5 20   8 Yes (0.40000000 0.60000000) *
         35) Price< 109.5 21   6 Yes (0.28571429 0.71428571) *
      9) Advertising>=13.5 45  20 Yes (0.44444444 0.55555556)
       18) Age>=54.5 20   5 No (0.75000000 0.25000000) *
       19) Age< 54.5 25   5 Yes (0.20000000 0.80000000) *
    5) Price< 92.5 46  14 Yes (0.30434783 0.69565217)
     10) Income< 57 10   3 No (0.70000000 0.30000000) *
     11) Income>=57 36   7 Yes (0.19444444 0.80555556) *
  3) ShelveLoc=Good 85  19 Yes (0.22352941 0.77647059)
    6) Price>=142.5 12   3 No (0.75000000 0.25000000) *
    7) Price< 142.5 73  10 Yes (0.13698630 0.86301370) *
```

What do the numbers in the parentheses mean? What do the stars mean?

Solution

The numbers in the parenthesis is probability of being no and yes, with the probabilty of being no and being yes in that order. Stars refer to terminal nodes, the end points of the tree

Part c (Explanation: 0.5 pts)

```r
set.seed(2)
train <- sample(1:nrow(Carseats2), 200)
Carseats.test <- Carseats2[-train,]

tree.carseats <- rpart(High ~ . - Sales,
                       data = Carseats2,
                       subset = train,
                       control = rpart.control(xval = 10))
names(tree.carseats)
```

```
 [1] "frame"                "where"                "call"
 [4] "terms"                "cptable"              "method"
 [7] "parms"                "control"              "functions"
[10] "numresp"              "splits"               "csplit"
[13] "variable.importance"  "y"                    "ordered"
```

```
tree.pred <- predict(tree.carseats, Carseats.test, type = "class")
table(tree.pred, Carseats.test$High, dnn = c("Predicted", "Actual"))
```

```
         Actual
Predicted No  Yes
      No  97   25
      Yes 20   58
```

What does the argument `xval = 10` mean?

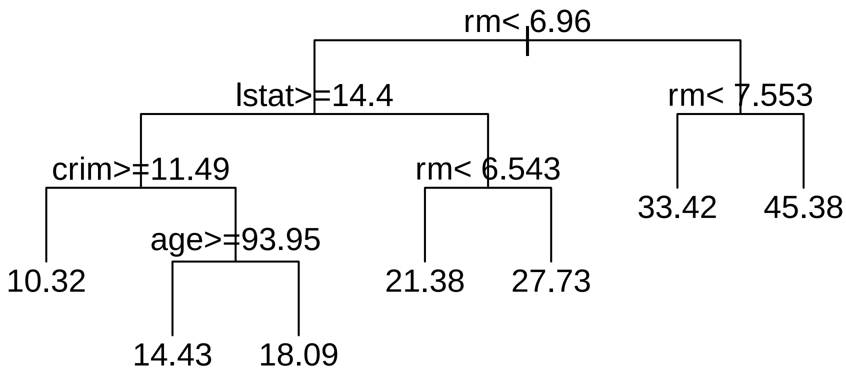
Solution

x

Part d (Explanation: 1 pt)

```
set.seed(1)
n <- nrow(Boston)
train <- sample(1:n, n/2)
tree.boston <- rpart(medv~., Boston, subset = train, control = rpart.control(xval = 10))
```

```
par(xpd = TRUE)
plot(tree.boston, uniform = TRUE)
text(tree.boston, pretty = 0)
```

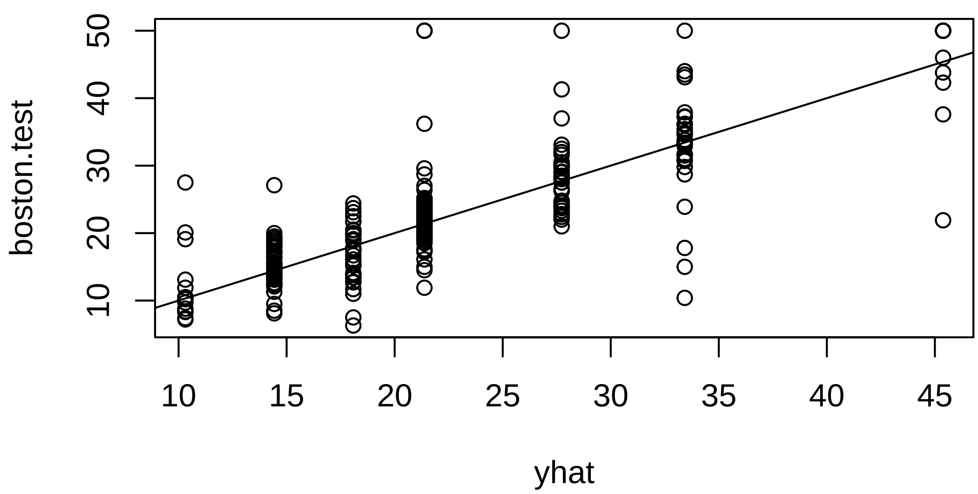


Briefly explain how to interpret this tree: which neighborhoods would be predicted to have a `medv` of 10.32? Which neighborhoods would be predicted to have a `medv` of 27.73?

Solution

Part e (Explanation: 1 pt)

```
yhat <- predict(tree.boston, Boston[-train,])
boston.test <- Boston$medv[-train]
plot(yhat, boston.test)
abline(0,1)
```




```
mean((yhat - boston.test)^2)
```

```
[1] 35.28688
```

Why do we get this pattern - a bunch of vertical lines - in the plot? (HINT: We saw plots like this in the example code from lecture, but they were horizontal lines because the predictions were on the verticalaxis instead.)

Solution

Part f (Code: 2.5 pts; Explanation: 1.5 pts)

We are now going to predict which public institutions are minority-serving institutions (HBCU or HSI).

Using tidymodels with the `rpart` package, fit a classification tree model on the training set. Tune the cost-complexity parameter using a grid of values between 0.01 and 1 with a reasonable enough number of levels so that you can actually see how your error metric of choice changes as a function of the cost-complexity parameter.

Choose a cost-complexity parameter by either Brier score or mean log-loss and fit that "best" model on the training set. Plot the final classification tree and explain how to use the tree to predict whether a school is MSI vs. non-MSI.

Solution

Part g (Code: 1 pt)

Using your final model from Part f, make predictions on the test set. Obtain the confusion matrix and compute the estimated Brier score or mean log-loss.

Solution

Problem 2: Random Forests and Boosted Trees

Part a (Code: 2 pts)

Run the code in ISLR Labs 8.3.3 and 8.3.4. Put each chunk from the textbook in its own chunk.

Solution

Part b (Code: 2.5 pts)

Using tidymodels, create a random forest model predicting whether an institution is a minority-serving institution (`MSI`) from the other variables in the dataset (except, obviously, `HBCU` and `HSI`, and `Name`). The `rand_forest` function will accept either `randomForest` or `ranger` as the `engine`.

Make sure to use `set_args` to tell tidymodels you want to use permutation-based importance (you will need it for part d). Tune the value of `mtry` using cross-validation; use values of `mtry` from 1 to 14 (14 = bagging).

Select your best model by cross-validated Brier score and justify your choice. Using your best model, obtain the out-of-bag prediction error (Brier score) on the training set.

Solution

Part c (Code: 1 pt; Explanation: 0.5 pts)

Using the model chosen in Part b, obtain predictions on the test set and compute the estimated test Brier score. Compare Brier score on the test set to the cross-validated mean Brier score and out-of-bag prediction error on the training set.

Solution

x

Part d (Code: 0.5 pts; Explanation: 1 pt)

Obtain and plot estimates of variable importance (`%IncMSE` column if you are using `randomForest` or `variable.importance` if you are using `ranger`) for the model chosen in Part b. Which predictor variables have the largest importance and are thus the most useful for making the predictions?

Solution

Part e (Code: 2.5 pts)

Using tidymodels with the `xgboost` package, fit a gradient-boosted tree model on the training set using the same predictors as the decision tree and random forest models. Tune only the 3 hyperparameters discussed in lecture: the number of trees to fit (`trees`), interaction depth (`tree_depth`), and learning rate (`learn_rate`). In order for this to not take forever, search over a small grid: 50, 100, 250, and 500 for `trees`, 1 and 2 for `tree_depth`, and 0.01 and 0.10 for `learn_rate`.

Select your best model by either Brier score or mean log-loss and justify your choice.

Solution

Part f (Code: 1 pt)

Using your best model from Part e, make predictions on the test set. Obtain the confusion matrix and the estimated Brier score or mean log-loss on the test set.

Solution

Part g (Code: 0.5 pts; Explanation: 1 pt)

Produce a variable importance plot for your xgboost model from Part e. Compare the (relative) variable importances between the random forest and xgboost models.

Solution

Part h (Explanation: 1 pt)

Consider the following statement: the second decision tree that is fit does not depend on the first decision tree that is fit. Is that statement TRUE or FALSE for the random forest algorithm? What about for boosted trees? Explain your reasoning.

Solution

Problem 3: Neural Networks

Part a (Computation and Explanation: 2 pts)

Consider two predictors, x_1 and x_2, feeding into a two-unit hidden layer. Suppose that the hidden layer uses ReLU activation functions with w_{kj} given in Equation (10.6) of the book and the output layer uses a linear activation function with β_j given in equation (10.6).

Find the activation functions $A_1(x_1, x_2)$ and $A_2(x_1, x_2)$ and $f(x_1, x_2)$. Note: $f(x_1, x_2)$ should be a piecewise function.

Solution

Part b (Compuation: 5 pts Extra Credit)

No.